AIORI 3 • ONLINE HACKATHON

NetworkFault AI

Evidence-Based Operator Network Fault Localisation

A proposed evidence-driven approach to investigating network faults through correlated measurements, transparent hypotheses and recommended verification tests.



PROBLEM STATEMENT PS-035	**TEAM** Cyberpolice	**INSTITUTE** Invertis University, Bareilly, Uttar Pradesh
TEAM MEMBERS Vivek Yadav • Priyanshu Ranjan	**FACULTY MENTOR** Mr. Mudit	**CATEGORY** Cybersecurity / Network Operations

PROJECT OVERVIEW

Abstract & Problem Description

NetworkFault AI is a proposed platform for evidence-based network fault localisation. It correlates network measurements and available logs to highlight suspected faults, show supporting evidence, and recommend verification tests. Optional AI-assisted explanations make findings easier to interpret.

Problem to Address

Difficult Fault Detection

Finding the device, link or segment behind latency, packet loss or connectivity failure can be challenging.

Scattered Evidence

Metrics, device status and logs may need to be examined across separate tools.

Time-Consuming Diagnosis

Manual troubleshooting can delay identification and verification of a likely fault.

Limited Transparency

An alert alone may not show why a particular device or link is suspected.

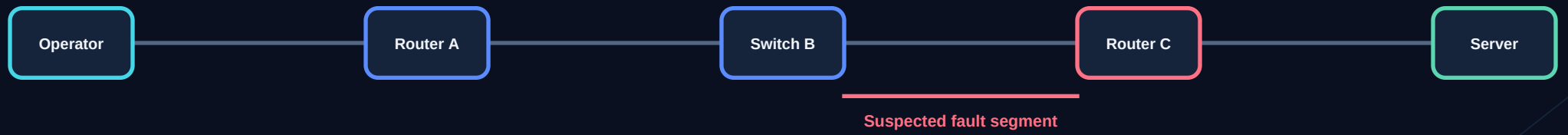
OBJECTIVE Develop an evidence-driven prototype that ranks potential fault locations and makes its reasoning and uncertainty visible.

TECHNICAL BACKGROUND

Background

Network fault localisation investigates which device, link or segment may be responsible for degraded connectivity or service availability.

Typical troubleshooting workflow



Illustrative topology; suspected location must be verified.

Existing Monitoring

Tools collect reachability, performance metrics and device events.

Operator Investigation

Operators compare symptoms across the network path and inspect logs.

Key Limitation

Evidence may be fragmented; one symptom can have multiple possible causes.

Need: A unified view that links observations to a suspected fault, communicates uncertainty, and suggests the next test needed for confirmation.

TECHNICAL REVIEW

Literature & RFC Review

Relevant protocols and tools provide the technical foundation for collecting and interpreting network evidence.

Tool / Standard	Relevance
Ping / ICMP	Checks reachability and observes round-trip response behaviour.
Traceroute	Helps inspect the path towards a destination; visibility may vary by network.
SNMP / Syslog	Can provide device counters, status and event messages when available.
RFC 792 — ICMP	Defines Internet Control Message Protocol.
RFC 1122 — Internet Hosts	Specifies requirements for Internet host communication.
RFC 1812 — IPv4 Routers	Specifies requirements for IPv4 routers.

Research Gap

Measurements and logs may be inspected separately. The proposed system focuses on correlating available signals and presenting a traceable hypothesis—not claiming certainty from a single alert.

Standards are technical references; they do not validate the proposed system or its performance.

SOLUTION

Proposed Solution

NetworkFault AI turns collected network observations into an explainable troubleshooting workflow.



Evidence-first diagnosis

Show observations that support each suspected fault.

Confidence & uncertainty

Communicate evidence strength and allow multiple hypotheses.

Verification guidance

Recommend a test to distinguish possible causes.

Unified dashboard

Present status, metrics, evidence and output together.

Expected outcome: A more structured and transparent investigation process. Effectiveness will be evaluated using controlled test scenarios.

OPTIMISATION

Solution Optimization

The proposed improvement is a more traceable troubleshooting workflow—not an unverified accuracy claim.

Dimension	Traditional workflow	Proposed NetworkFault AI
Evidence review	Inspect metrics and logs separately	Correlate available observations in one workflow
Fault localisation	Build a hypothesis manually	Rank suspected location with supporting evidence
Explainability	Interpret alerts and raw output	Show why a location is suspected and what remains uncertain
Next action	Choose follow-up checks manually	Recommend a verification test
Review	Compare results across tools	Keep findings in a unified dashboard

Evaluation Plan

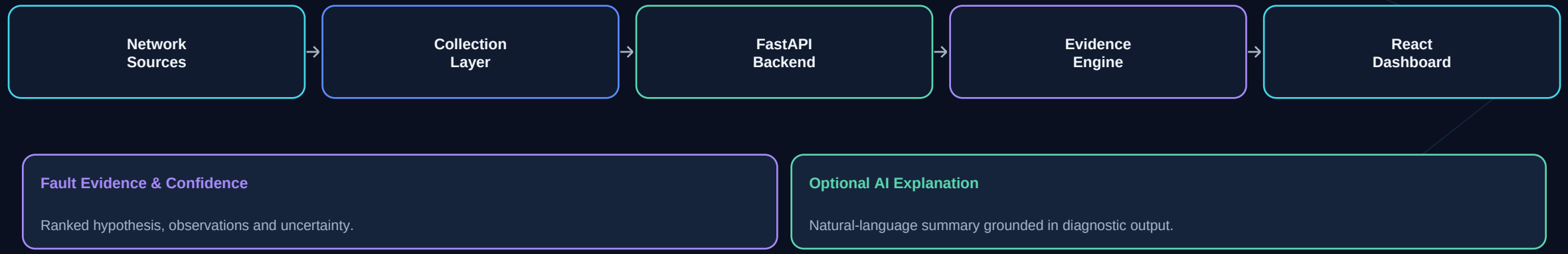
Test against known healthy and controlled-fault scenarios. Compare the suspected location with the known injected fault, record false positives, and verify whether the recommended next test is useful.

No numerical improvement is claimed before testing.

SYSTEM DESIGN

Solution Architecture & Design

Proposed modular architecture for collecting evidence, analysing it and presenting diagnostic findings.



Output: suspected fault • evidence • uncertainty • recommended verification

Technology Stack

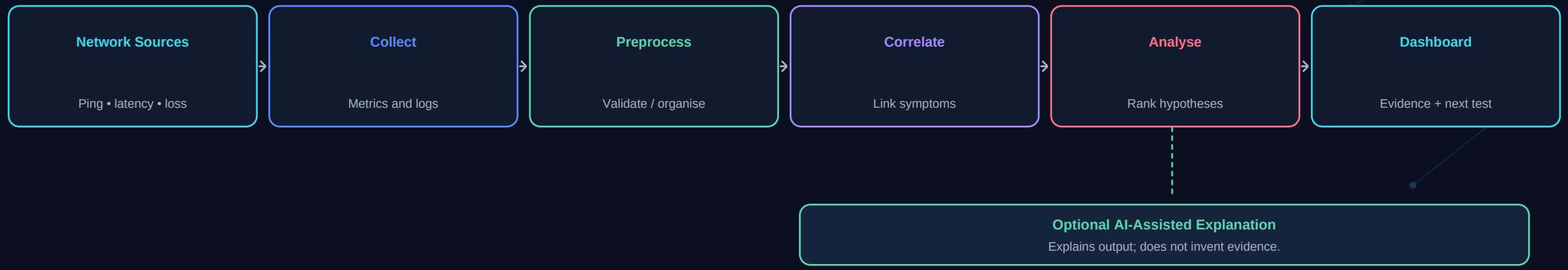
Frontend	Backend	Network evidence	Storage / AI
React • Vite • Tailwind CSS	Python • FastAPI	Ping • Traceroute • Metrics	SQLite/PostgreSQL • optional LLM

Optional components will be included only where implementation and test data support them.

DATA PIPELINE

Data Flow Diagram

How measurements move through the system and become a testable fault hypothesis.



OUTPUT Suspected fault • Supporting evidence • Confidence / uncertainty • Recommended verification test

Inputs require authorised access or controlled test data. The optional LLM explains diagnostic output; the evidence engine remains responsible for the underlying hypothesis.

IMPLEMENTATION PLAN

Deployment Timeline

Proposed milestones; sequence and scope may be adjusted as implementation progresses.



Proposed sequence; milestones will be updated as implementation progresses.

Validation loop

Healthy baseline → controlled fault → collect evidence → compare diagnosis with known fault → restore and re-test

SOURCES

Bibliography

Technical references relevant to the proposed prototype.

RFC 792	Internet Control Message Protocol (ICMP). RFC Editor.
RFC 1122	Requirements for Internet Hosts — Communication Layers. RFC Editor.
RFC 1812	Requirements for IPv4 Routers. RFC Editor.
React	React documentation — react.dev
FastAPI	FastAPI documentation — fastapi.tiangolo.com
Python	Python documentation — docs.python.org/3/
Ping / Traceroute	Standard network diagnostic utilities; platform-specific behaviour may vary.

References describe underlying protocols and tools; the proposed system's results must be validated independently.

CLOSING

THANK YOU

NetworkFault AI

Turning network evidence into actionable insights.



TEAM CYBERPOLICE	FACULTY MENTOR	INSTITUTE
Vivek Yadav • Priyanshu Ranjan	Mr. Mudit	Invertis University, Bareilly

Evidence first. Hypotheses transparent. Verification always.